CS-2008 Numerical Computing Quiz 3 - BCS-7A Fall 2025

Name:_____ Roll Number: _____

1. Consider the integral $I = \int_0^2 (x^3 + 1)dx$. What is the approximation of I using the basic **Trapezoidal Rule** ($n = 1$)?

A. 4

B. 10

C. 6

D. 5

2. For the integral $I = \int_0^2 (x^3 + 1)dx$, what is the approximation of I using the basic **Midpoint Rule** ($n = 1$)?

A. 6

B. 10

C. 5.25

D. 4

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3. Calculate the approximation of $I = \int_0^2 (x^3 + 1)dx$ using the basic **Simpson's $1/3$ Rule.** (Note: The basic rule requires $n = 2$ intervals).

A. 7.5

B. 6

C. 6.25

D. 5

4. What is the approximation of $I = \int_0^2 (x^3 + 1)dx$ using the basic **Simpson's $3/8$ Rule?** (Note: The basic rule requires $n = 3$ intervals).

A. 6.125

B. 4.8

C. 5.5

D. 6

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5. Use the **Composite Trapezoidal Rule** with $n = 4$ intervals to approximate $I = \int_0^2 (x^3 + 1)dx$

A. 6.25

B. 6.0625

C. 6

D. 5.875

6. Use the **Composite Midpoint Rule** with $n = 4$ intervals to approximate $I = \int_0^2 (x^3 + 1)dx$.

A. 5.875

B. 5.75

C. 6.25

D. 6

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7. Use the **Composite Simpson's $1/3$ Rule** with $n = 4$ intervals to approximate $I = \int_0^2 (x^3 + 1)dx$.

A. 6

B. 6.25

C. 6.125

D. 5.9

8. Use the **Composite Simpson's $3/8$ Rule** with $n = 6$ intervals to approximate $I = \int_0^2 (x^3 + 1)dx$.

A. 6.05

B. 6.25

C. 5.9375

D. 6

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9. Which of the following numerical integration rules **requires** the number of intervals (n) to be a multiple of 3?

A. Trapezoidal Rule

B. Midpoint Rule

C. Simpson's $1/3$ Rule

D. Simpson's $3/8$ Rule

10. Given the Composite Trapezoidal Rule result $T_4 = 6.25$ and the Composite Midpoint Rule result $M_4 = 5.875$ (for $I = \int_0^2 (x^3 + 1)dx$ with $n = 4$, exact value is 6), which of the following is true regarding their absolute error?

A. The Midpoint Rule overestimates the integral, while the Trapezoidal Rule underestimates it.

B. The Trapezoidal Rule error is twice that of the Midpoint Rule error.

C. The Midpoint Rule error is twice that of the Trapezoidal Rule error.

D. Both rules have the same absolute error.

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Formulas

$$I \approx \text{Midpoint Rule } (M_1) = (b - a) \cdot f\left(\frac{a+b}{2}\right)$$

$$I \approx \sum_{i=1}^{n} f(\bar{x}_i)\,\Delta x$$

$$I \approx \frac{b-a}{2}[f(a) + f(b)]$$

$$I \approx \frac{b-a}{2n}[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)]$$

$$I = \frac{h}{3}(f_0 + 4f_1 + f_2)$$

$$\frac{1}{3}h\left[f(x_0) + 4\sum_{i=1}^{n/2} f(x_{2i-1}) + 2\sum_{i=1}^{n/2-1} f(x_{2i}) + f(x_n)\right]$$

$$\int_a^b f(x)\,dx \approx \frac{3h}{8}\left[f_0 + 3f_1 + 3f_2 + f_3\right]$$

$$\int_a^b f(x)\,dx \approx \frac{3h}{8}\left[f_0 + 3\sum_{\substack{1\le i\le n-1 \\ i\not\equiv 0 \pmod 3}} f_i + 2\sum_{\substack{3\le i\le n-3 \\ i\equiv 0 \pmod 3}} f_i + f_n\right], \quad n = 3m$$

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